Def.

A space $X$ is called a Kolmogorov (or $T_0$) space if: given $x, y \in X$ with $x \neq y$, there exists an open set $U \subseteq X$ such that either $\begin{cases} x \in X \\ y \notin X \end{cases}$ or $\begin{cases} x \notin X \\ y \in Y \end{cases}$.

Prop.

$X$ is $T_0$ space if and only if $x \neq y \Rightarrow \overline{\{x\}} \neq \overline{\{y\}}$

Def.

A space $X$ is called a Fréchet space (or $T_1$) iff given $x, y \in X$ with $x \neq y$, there exists open sets $\mathcal{U}, V \subseteq X$ such that $\begin{cases} x \in \mathcal{U} \\ y \notin \mathcal{U} \end{cases}$ and $\begin{cases} x \notin V \\ y \in V \end{cases}$

Prop.

$X$ is $T_1$-space if and only if for all $x \in X$, $\overline{\{x\}} = \{x\}$.

Proof. Suppose that $X$ is $T_1$-space. Let $x \in X$ be fixed. Given $y \in X \setminus \{x\}$, there exists an open $\mathcal{U}_y \subseteq X$ such that $x \notin \mathcal{U}_y$ and $y \in \mathcal{U}_y$. So,

$$\bigcup_{y \in X \setminus \{x\}} \mathcal{U}_y = X \setminus \{x\}, \text{ because } \bigcup_{y \in X \setminus \{x\}} \mathcal{U}_y \supseteq X \setminus \{x\} \text{ but it does not contain } x.$$

Now, since union of open sets is open, the complement $\{x\}$ is closed set.

By the definition of closure, we obtain $\overline{\{x\}} = \{x\}$.

Conversely, let $x, y \in X$ be distinct points. Then,

$$\begin{cases} x \in X \setminus \{y\} \\ y \notin X \setminus \{y\} \end{cases} \text{ and } \begin{cases} x \notin X \setminus \{x\} \\ y \in X \setminus \{y\} \end{cases}, \quad \text{So Proved } \boxtimes$$

Note: By definition, trivially $T_2 \Rightarrow T_1 \Rightarrow T_0$.